ATTAIN FY25–26 · IOWA DOT, LEAD APPLICANT · CCAI · I–35 & I–80 CORRIDORS · BRIEFING EXHIBIT

Trusted work zone and truck parking data across I-35 and I-80

Sixteen states keep publishing exactly what they publish today. The project brings **public and private data together**, verifies every work zone, predicts parking at every facility, and delivers one trusted feed to every navigation app, truck, and connected vehicle — border to border on I-35, coast to coast on I-80, meeting at the Iowa crossroads — so a driver ending a shift finds a legal, safe parking spot in **30 minutes instead of 60**.

How the data flows

Read top to bottom: what exists today enters at Layer 1; drivers receive the result at Layer 5.

LAYER 1

Public data the sixteen states already produce — unchanged

No new systems, no new staff, no new publishing obligations. Participation and support of the shared feed is the entire ask. Where a state has no digital feed today, proven lightweight tooling generates one from data the state already produces (demonstrated for Nebraska and Nevada).

Work zones

Work zone & incident reports from each state's existing 511 / traffic management systems

Truck parking

TPIMS space-detection at instrumented facilities · ground-truth devices (cameras, in-ground sensors) — incl. Iowa rest-area cameras with AI occupancy counts, live today · historical occupancy archives that teach the predictions what "normal" looks like

I-35 · Texas · Oklahoma · Kansas · Missouri · Minnesota · I-80 · California · Nevada · Utah · Wyoming · Nebraska · Illinois · Indiana · Ohio · Pennsylvania · New Jersey · Iowa — where the corridors meet

▼ joined by

LAYER 2

Private & partner data from the road itself

Independent signals confirm what's really happening — whether each reported work zone is truly active, and how full each parking facility actually is.

Verifies work zones

Vehicle probe & movement data · construction e-ticketing (material deliveries) · connected field devices (arrow boards, cones, sensors) · connected-truck in-cab alert networks

Sharpens parking predictions

Fleet telematics & ELD provider input (where trucks actually park and dwell) · crowd-sourced driver apps · connected-vehicle movement analytics · camera-based AI counts · closure & work zone activity that will consume downstream parking

▼ brought together in

LAYER 3

The verification & prediction engine

Where it all comes together. Every work zone is cross-checked against the evidence and given a **confidence score**; map errors are corrected to follow the actual roadway. Every parking facility gets an **hourly occupancy prediction** — trained on historical occupancy, fed by every live signal, continuously calibrated against ground-truth counts. When a closure will fill the next three truck stops, the engine knows **before the driver's shift ends**.

- Confidence score on every event
- Hourly parking prediction per facility
- Closure → parking impact linkage
- Monthly state report cards

▼ exchanged through

LAYER 4

One governed exchange — MITRE iNODE

A single, secure connection point with provenance and access control. Consumers connect **once** to reach all sixteen states — instead of negotiating fifty separate state integrations. The exchange manages the flow **in both directions**: validated data out to consumers, and verified corrections and field evidence back to inform each state's source feed.

▼ delivered to

LAYER 5

The people who need it, in the systems they already use

Work zones published in the two formats the market consumes — WZDx (federal standard) and CIFS (the format Google and Waze ingest directly). Parking availability and predictions delivered to the driver's cab, the dispatcher's screen, and roadside signs.

- Navigation apps (Google · Waze · Apple)
- Trucks & fleets (in-cab alert networks · telematics)
- Connected & automated vehicles (V2X / C-V2X message distribution)
- DMS roadside signs
- Public dashboard & FHWA

60 → 30 min

END-OF-SHIFT PARKING SEARCH

Cutting search time in half pays out four ways, every day, corridor-wide:

- Supply chain productivity ↑
- Fuel burned searching ↓
- Trucks on ramps & shoulders ↓
- Hours-of-service violations ↓

RETURN LOOP

Did the driver see it? Did the source feed learn?

Two loops close through the MITRE-managed exchange. **Market penetration:** the observatory compares what states publish against what navigation services actually display to travelers (probe-data providers today; the major consumer map platforms by invitation) — so leadership can see which operational practices get work zones in front of drivers, and standardize what works across all sixteen states. **Feedback to the source:** validated field evidence and verified corrections return through the exchange to inform each state's original feed — the system continuously improves the data it started with.

Why this architecture wins

Almost nothing is asked of the states

The layered design means states contribute what they already produce. Quality is raised above them, not demanded from them — so sixteen states can say yes quickly, and more can join later.

Trust, measured in public

Every event carries a confidence score; every state gets a monthly report card on a public dashboard. FHWA can watch corridor data quality improve for the life of the award.

One connection instead of fifty

Google, Waze, fleets, and AV developers connect once to the exchange and receive the whole corridor — the adoption barrier that has held work zone data back simply disappears.

Built on systems running today

The ingestion platform, quality scoring, and probe comparison are operational now across roughly 70 data sources and 39 work zone feeds. Grant funds deployment, not invention.

What the grant buys

$6.0M Federal (71% share) · $2.4M non-Federal committed (29% — exceeds the 20% required; led by private-partner in-kind) · $8.4M total · 36 months + evaluation period. Right-sized to the program: $120M across ~20 expected awards.

DEPLOYMENT ELEMENT	FEDERAL
Private data subscriptions & partner data agreements (telematics/ELD, probe & CV analytics, e-ticketing, in-cab alerts)	**$1.4M**
Truck parking availability & prediction network (detection at priority gap facilities, delivery to cab/dispatch/signs)	**$1.0M**
Connected work zone field pilot (~12–15 zones per season, automated device-to-event linkage)	**$1.0M**
Standardized publication & geometry correction across all 16 corridor states	**$0.7M**
Verification & prediction engine, public dashboard & evaluation	**$0.7M**
Exchange backbone — governed multi-state exchange + V2X/C-V2X distribution	**$0.6M**
Program management, systems engineering, governance	**$0.4M**
Outreach, training & national transferability toolkit	**$0.2M**
Total Federal request	**$6.0M**

Iowa DOT, lead applicant · supporting CCAI year-one work plan · I-35 Anchor State · I-80 Corridor Coalition · partner states join by letter of commitment · Prepared as a leadership briefing exhibit for the FHWA ATTAIN application (FY25–26 NOFO · 20% non-Federal match · deadline August 11, 2026). Baseline: ATRI driver-diary research — drivers average ~56 minutes/day searching for parking, ≈$5,000/year in lost wages. Private-sector data categories are shown generically for procurement neutrality; the market includes probe/movement analytics, construction e-ticketing, connected field devices and ground-truth sensors, in-cab alert networks, fleet telematics and ELD providers, and crowd-sourced driver apps — selected competitively after award. V2X/TIM message distribution services are likewise procured competitively.